

Ultrasound-guided Robotic Aspirator for Autonomous Thoracentesis in Pleural Effusion Treatment

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Abstract—Thoracentesis is a crucial yet challenging procedure for treating pleural effusion, a condition characterized by the accumulation of excess fluid in the pleural space. Despite its routine use, the procedure demands precise needle placement to avoid complications such as pneumothorax or hemorrhage. This paper presents a novel ultrasound-guided robotic system designed to automate thoracentesis, enhancing precision and reducing the risk of human error. The proposed system integrates a custom robotic end-effector with a Franka Emika Panda robot, combining real-time ultrasound imaging, needle insertion, and fluid aspiration into a compact and cost-effective solution. Extensive testing on a pleural phantom simulator demonstrated high accuracy, improved precision, and reduced procedure time. This work offers a significant advancement in the safe and efficient treatment of pleural effusion.

Index Terms—needle insertion system, autonomous thoracentesis, pleural effusion, ultrasound-guided operations.

I. INTRODUCTION

Pleural diseases, such as pleural effusion, affect 360 per 100,000 people and are expected to increase due to population growth [1]. Lung cancer, a leading cause of pleural effusion, often requires prompt fluid drainage [2]. In 2022, lung-related cancers caused 1.8 million deaths, with cases projected to rise to 3 million by 2045 [3]. Pleural effusion occurs when fluid accumulates in the pleural space, impairing breathing. Normally, this space contains about 0.2 mL of fluid to enable smooth lung movement during respiration [4], [5]. In severe cases, fluid buildup can reach up to 2 liters, leading to respiratory distress and serious health issues.

Thoracentesis is a common minimally invasive procedure employed to treat pleural effusion by draining the excess fluid. Despite its routine nature, thoracentesis remains technically challenging due to the narrow and delicate nature of the pleural space, requiring precise needle placement to avoid complications like pneumothorax (air in the pleural cavity) or hemorrhage [6]. To minimize risks, advanced imaging techniques such as Computed Tomography (CT) scans, X-rays, and ultrasounds are often employed [7]. However, using auscultation without imaging increases error risk, potentially harming surrounding tissues. The success of thoracentesis relies heavily on clinician expertise, with factors like hand tremors or fatigue further affecting precision.

For imaging in thoracentesis, several techniques can be used. Chest X-ray (CXR) and CT both rely on ionizing radiation, with CT offering superior image quality and the

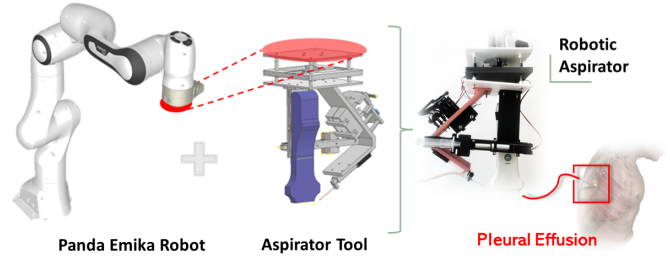


Fig. 1: Robotic aspirator for pleural effusion treatment: conceptual scheme.

ability to distinguish between different tissue layers, though it is more expensive and complex [8]. CXR is effective for detecting large fluid volumes but requires at least 200 mL for reliable detection [9]. In contrast, transudative Ultrasound (US) provides a non-invasive alternative with high diagnostic accuracy, detecting fluid with 94.67% accuracy compared to 65.33% for CXR [10]. US uses sound waves to visualize internal structures, allowing for the detection of smaller fluid volumes and offering real-time guidance during thoracentesis, which reduces the risk of accidental puncture of nearby organs or vessels [11], [12], [13]. Its portability, ease of use, and effectiveness in guiding thoracentesis and monitoring fluid drainage make US a valuable tool [14].

Recent advancements in robotic surgery have significantly enhanced the precision, control, and safety of various medical procedures, particularly in minimally invasive interventions [15] by being adapted for tasks which require high accuracy, such as percutaneous procedure [16] or for drug administration [17] to deliver medicine to specific organs due to their ability to perform consistent and precise movements while minimizing human error. In particular, robotic injection systems have shown great promise in improving outcomes for procedures like biopsies [18], brachytherapy [19], and ablations [20] by offering real-time imaging integration and autonomous needle guidance [21], [22].

These systems are designed to reduce complications associated with manual needle manipulation [23], such as tissue damage or inaccurate placement, making them ideal for complex thoracic procedures like thoracentesis. Recently, Anxing Jiang et al in [24] proposed an automated system for small-scale pleural effusion aspiration procedures. However, to the best of our knowledge, up to now, no fully autonomous robotic system currently exists that performs thoracentesis under real-time US guidance.

Additionally, the integration of advanced imaging tech-

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nologies, particularly US, with robotic platforms significantly improves the precision of targeting and treating affected areas [25], [26]. This synergy between imaging and robotics enhances the safety and effectiveness of minimally invasive procedures, highlighting the importance of robotic-assisted needle insertion for pleural effusion treatment [27]. These advancements support the development of our fully autonomous, US-guided robotic thoracentesis system, which aims to set a new standard in this critical medical procedure.

This paper presents a fully autonomous, ultrasound-guided robotic system designed to perform thoracentesis, addressing the challenges of precision and efficiency in pleural effusion treatment (Figure 1). Key contributions include also the development of a custom robotic end-effector that integrates essential subsystems for thoracentesis: real-time ultrasound scanning, a Needle Insertion Module (NIM), and a Fluid Aspiration and Drainage Module (FADM). These components are coupled with electronics for autonomous control. The system, implemented on a Franka Emika Panda robot, operates seamlessly without human intervention, offering a cost-effective, compact, and modular solution that can be adapted to various robotic platforms, enhancing the surgeon's capabilities to perform the procedure. Extensive experiments conducted on a pleural simulator phantom demonstrate that the proposed robotic system can perform thoracentesis with a high degree of accuracy, significantly improving both precision and procedure time.

II. ROBOTIC END-EFFECTOR - ASPIRATOR

This section outlines the custom-designed physical components developed for the robotic aspirator. Central to this work is the design and fabrication of a specialized end-effector (EE), created using 3D printing to allow precise customization for the medical instruments involved. Designed in Autodesk Inventor, the aspirator is securely attached to the robot's end-effector, consolidating into a single structure that facilitates both needle insertion and fluid aspiration. The aspirator's dual-purpose design ensures precise US probe placement for needle guidance, while the independent aspiration system extracts pleural fluid via plastic tubing. This modular approach enhances both the precision and functionality of the thoracentesis procedure, ensuring optimal performance in clinical environments.

A. Mechanical design of the end-effector

The base structure of 14x14cm, mounted to the end effector of a Panda robot, serves as both an electronics enclosure for the Arduino board and actuator driver, and as the primary support for the system's subsystems. Figure 2 shows the CAD model of the end-effector, highlighting its key subcomponents. The NIM Holder has an "L-shape" with a flattened angle (in red in Figure 2), where its longer leg connects to the base at a 45-degree angle, ensuring the rotation angle of the needle insertion. The needle insertion mechanism is mounted on the outer side of the shorter leg, where the needle actuator of 50mm stroke length is secured by a locking structure. A 23G needle with 0.64mm diameter

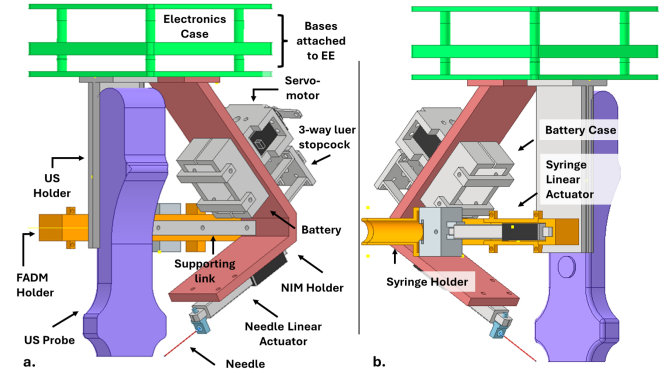


Fig. 2: CAD model of the robotic aspirator: (a) front view showing probe holder and NIM components; (b) rear view showing fluid aspiration and syringe holder. Highlighted in colors: EE base (green) houses electronics, US probe (purple), NIM holder (red), and syringe holder (orange).

and 38mm length, is connected to the tip of this actuator through a custom adapter, points downward toward the body, allowing precise insertion directly beneath the ultrasound scanner (as shown in Figure 2). The actuator extends to its maximum to insert the needle to the target area and retracts upon completion. For added stability, a supporting link is attached to the inner side of the NIM Holder, with other components mounted on both the inner and outer sides of the holder (shown as orange and grey in Figure 2).

US Holder. The 5x14 cm ultrasound (US) scanner holder is mounted perpendicularly to the base and the supporting link. Its design secures the Sonon 300L US attachment, which measures 23 cm in length, 8 cm in height, and has a 4x1 cm contact area. A Velcro mechanism allows for easy removal and reattachment of the scanner. Its vertical orientation ensures optimal image acquisition by maintaining the correct alignment with the target area.

Aspiration and Expulsion Mechanism. The FADM holder supports both a syringe of 50mL volume and a linear actuator with 30mm stroke. The syringe is firmly fixed to the holder, which is mounted along the supporting link, with the Syringe side positioned closer to the NIM Holder. The Syringe Linear Actuator is responsible for manipulating the syringe plunger, with the help of a slitted cuboid shaped adaptor, enabling the aspiration and expulsion of fluids. This is achieved by retracting the linear actuator's rod when aspirating and extending it when the fluid needs to be expelled.

Valve System. The valve system manages the switch between aspiration and expulsion functions through a 3-way luer stopcock, which connects the syringe, needle, and expulsion tubes of 7 mm in diameter. The 3-way luer stopcock is mounted on a holder which is attached to the outer side of the longer leg of the main frame, near the flattened angle (see 2) and is fixed in place by two arches, allowing only the handle to move. The valve handle, located on top of the tap, is fitted with an adapter that enables rotation via a servo-motor.

Robotic US scanning

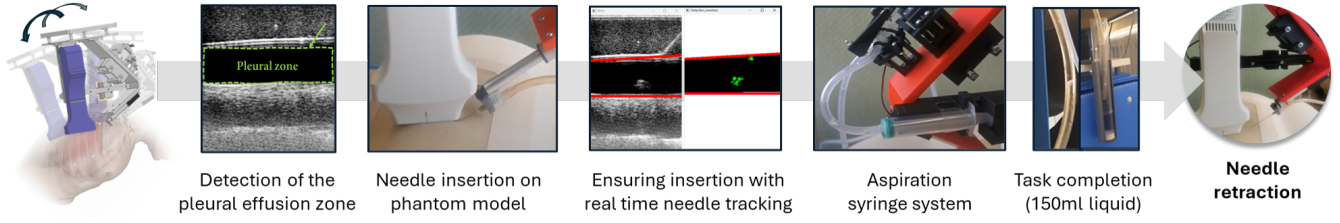


Fig. 3: General process of thoracentesis; schematic overview.

The servo-motor (DSS-M15S/DFROBOT), fixed in a case on the NIM holder near the base, controls the valve. Its double horn is linked to the valve adapter through two piston-like links, rotating the valve to direct fluid flow. When the valve points toward the expulsion tube, it closes the tap to enable aspiration. After aspiration is complete, the servo rotates the valve to close the aspiration tube and allow expulsion. Opposite the servo, a battery case is secured to the inner side of the NIM Holder, with two locking structures ensuring the 9V battery remains stable during robot movement.

B. Electronic components

For ensuring precise control, communication, and execution of mechanical operations in both NIM and FADM modules, several electronic components are integrated into the end effector. The Arduino Uno serves as the primary control interface, handling serial communication with ROS for closed-loop control, and managing both linear actuators and a servo motor. The Arduino controls the 12V 50mm stroke linear actuator for needle insertion and retraction in NIM, and the 6V 30mm stroke actuator for syringe plunger movement in FADM, enabling fluid extraction and expulsion. A dual H-bridge motor driver provides bidirectional control for both actuators, ensuring precise movements. Additionally, the Arduino manages a servo motor that rotates the 3-way luer cockstop, allowing for automated switching between fluid pathways during aspiration. This servo motor ensures accurate valve positioning, preventing fluid leakage or back-flow, thereby enhancing the overall precision and safety of the thoracentesis procedure.

III. ROBOT-ASSISTED THORACENTESIS

The autonomous thoracentesis process begins with the *US Guidance and Monitoring* phase, where the robotic EE, holding the US scanner, approaches the thoracentesis pad as shown in Figure 3. A control command is sent from the master PC, positioning the EE precisely above the phantom model. Upon reaching the target position, the system confirms this and captures ultrasound images. These images are then processed to identify the pleural zone for accurate fluid aspiration. Following this, the *Needle Insertion Module* is activated to insert the needle into the pleural space, with real-time ultrasound monitoring ensuring precise placement. Once the needle reaches the target area, the *Fluid Aspiration and Drainage Module* begins fluid extraction, continuing until

100 to 150mL of fluid is removed. After extraction, the NIM retracts the needle, and the robotic system moves away for post-operative monitoring. In the following, detailed explanations of the subprocedures will be provided, including Robot Control for Autonomous US Scanning, Real-Time Detection of Pleural Effusion Zone, Needle Insertion Module, Needle Tracking, and Fluid Aspiration and Drainage Module.

A. Robot control for autonomous US scanning

The autonomous scanning process is algorithmically managed through ROS and MoveIt to ensure precise control over the scanning area on the phantom model. The scanning methodology presupposes that the phantom is fixed at a known position. Then five key points on the phantom surface are identified, as described in [16] and arranged in a circular pattern with a 2mm radius. These key points define the area to be scanned.

Then MoveIt generates the necessary trajectories to cover the defined scanning area. This planning ensures that the robot's movements are both efficient and collision-free, considering the robot's kinematic constraints and workspace boundaries. Once the trajectories are computed, the robot executes them using the Franka impedance controller, moving the US scanner across the target area in a precise manner. This approach allows for comprehensive coverage of the scanning region, ensuring high-quality and accurate US imaging. ROS facilitates real-time communication and control of the robot, while MoveIt ensures that the planned paths are executed smoothly, resulting in effective and efficient scanning.

B. Real-time detection of pleural effusion zone

The US scanner transmits images directly to an android tablet over a wireless connection. To integrate the US images into the robotic control system, the real-time images need to be accessible on a PC, for further image processing. This is achieved using the tool screpy (Screen Copy), which allowed real-time streaming of the US images from the tablet to the PC with minimal latency. Once the US images are mirrored on the PC, they can be processed in real time to detect the pleural effusion zone. The detection of the pleural zone, or target position (TP), relies on a real-time processing algorithm of the US images. Since the US data is captured as a continuous video, the system processes the first frame to create a binary mask using a combination of morphological

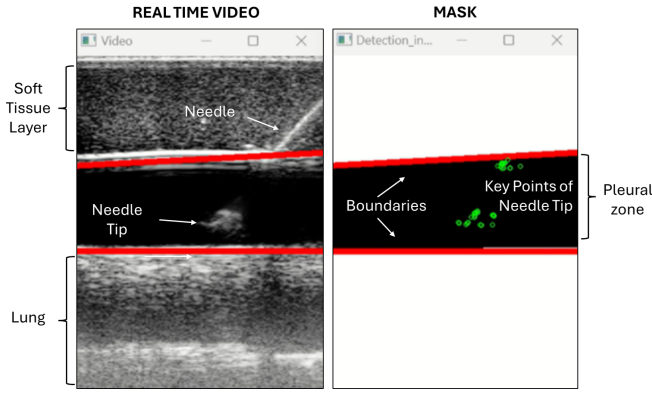


Fig. 4: Snapshot US image during scanning (left) and the corresponding mask of the pleural zone (right). Detection of needle tip in US scan through key points (green dots).

operations, such as dilation and erosion that highlight the relevant areas and features. Thereafter, this mask is enhanced by using a line detection algorithm based on Canny Filter and then fine-tuned with Hough Transform to identify the boundaries of the pleural zone by detecting the edges of the tissue layers, i.e. ribs and organs.

Figure 4 shows a frame from the US data collected for processing and the corresponding mask on the right window. The boundaries of the pleural zone, shown in red, are detected real time and then the mask is created. This mask is critical for processing video frames, guiding the activation of the subsystems, and tracking the needle tip during the NIM module's operation.

C. Needle Insertion Module

Once the pleural effusion zone is identified, the robot initiates the needle insertion process through coordinated communication between ROS and the external hardware comprising of Arduino and other electronics components. The robot's control system within ROS first sends a command via serial communication to Arduino to extend the linear actuator of the NIM by 50mm. This actuator, to which the needle is attached, then moves the needle into the pleural space. Arduino drives the actuator with precision, ensuring controlled needle insertion and minimizing tissue damage by calibrating the actuator's speed and extension distance. After the needle reaches the target area, Arduino sends a confirmation message back to ROS, completing the feedback loop and ensuring that the robot's control system is aware of the insertion status before proceeding to the next phase. At the same time, the monitoring system ensures needle insertion with the real-time needle tracking algorithm.

D. Needle Tracking

The needle tip is tracked using the Oriented FAST and Rotated BRIEF (ORB) algorithm [28], which combines keypoint detection and descriptor extraction to effectively monitor the needle's position. ORB first uses the FAST keypoint detector to identify corners or distinct features within the image, then applies the BRIEF descriptor, which

encodes the appearance of these keypoints into a binary string. This process allows ORB to detect and generate key points (KPs) within the targeted zone, highlighted in green in Figure 4. As the needle moves during insertion or removal, the algorithm continuously tracks these KPs, and their transitions indicate needle movement. Specifically, if the number of detected KPs exceeds a threshold of 25 (experimentally determined), it confirms that the needle is being inserted, while a decrease below this threshold signals needle retraction.

E. Fluid Aspiration and Drainage Module

After successful needle insertion into the pleural effusion zone, the fluid aspiration process is initiated through coordinated control by ROS, Arduino, and the FADM components. The procedure is as follows:

- Initiating Syringe Retraction for Fluid Extraction.

Once ROS confirms from Arduino that the needle is in the target area, it commands Arduino to begin fluid extraction. Arduino activates the linear actuator connected to the syringe plunger, retracting it by 30mm to create negative pressure that draws fluid from the pleural zone into the syringe. During this step, the luer cockstop valve blocks fluid flow towards the drainage bottle, ensuring that fluid flows solely from the needle into the syringe.

- Fluid Collection in the Syringe.

As the syringe fills with fluid, Arduino monitors the actuator's position to control the volume extracted. Once the target volume is reached, Arduino sends a confirmation message back to ROS indicating that the fluid collection is completed.

- Switching Valve Position with Servo Motor.

Upon receiving the confirmation from Arduino, ROS sends a command to rotate the luer cockstop handle. Arduino activates a servo motor connected to the luer handle via a 3-way link mechanism, rotating the handle 90 degrees. This rotation repositions the luer handle to block the valve towards the needle and unlock the valve towards the drainage bottle, as indicated in Figure 5.

- Syringe Plunger Forward Movement to Drain Fluid.

After the luer handle rotation is confirmed by Arduino, ROS sends another message to Arduino to initiate the forward movement of the syringe plunger. The linear actuator pushes the plunger forward, expelling the collected fluid through the tube into the drainage bottle.

- Repositioning Luer Handle to Initial State:.

Once the fluid is successfully transferred, Arduino sends a completion message to ROS. ROS then commands Arduino to rotate the luer handle back 90 degrees anticlockwise, returning it to the initial position and the system is read to start the next cycle.

The aspiration cycle is repeated three times to achieve a total extraction of approximately 150ml of fluid from the phantom. Each cycle involves the same sequence of steps: syringe retraction, fluid collection, valve repositioning, fluid expulsion, and handle resetting. ROS-Arduino communication manages each transition to ensure that every phase is synchronized, reducing risks of errors during the repetitive aspiration process. After the final aspiration cycle, Arduino

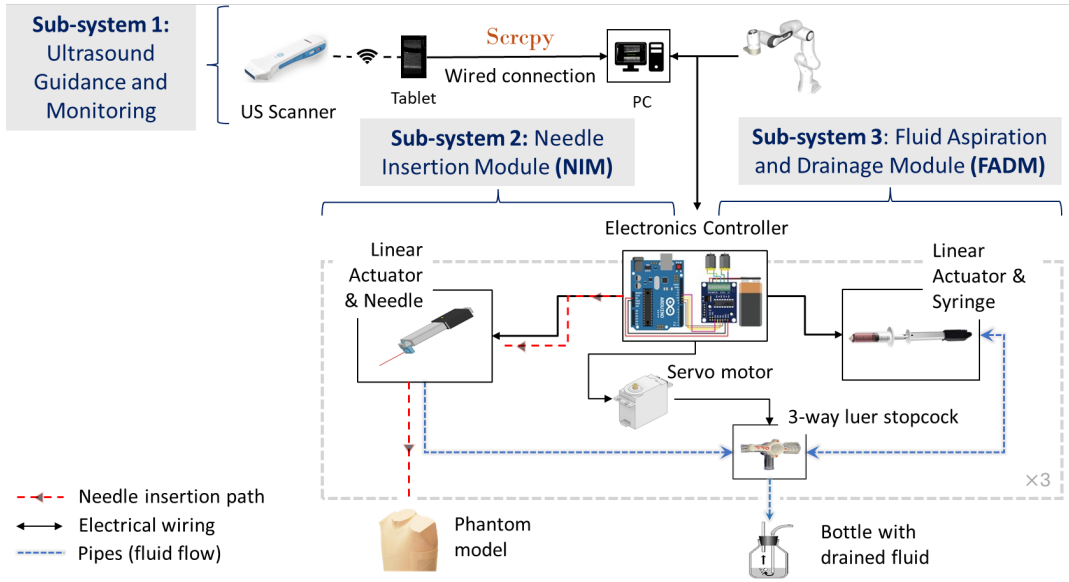


Fig. 5: Detailed workflow diagram of the sub-systems, namely US guidance and monitoring, NIM and FADM, with the electronic connections (black line), the needle insertion path (red dotted line) and the fluid flow (blue dotted line).

sends a confirmation message to ROS indicating that the aspiration process is completed. ROS then sends a final command to Arduino to initiate needle retraction.

IV. EXPERIMENTAL RESULTS

A. Experimental Setup

A realistic laboratory scenario was established to experimentally assess the developed autonomous thoracentesis system, as illustrated in Figure 6. The experiments were conducted using an US-guided thoracentesis simulator by Kyoto Kagaku, Japan, which is designed to replicate real-life thoracentesis interventions. This simulator includes two puncture units: one for mid-scapular line access and another for mid-axillary line access. Figure 7 displays the phantom simulator. On the left is the external view of the torso manikin and the scapular line access pad, measuring W16 x D7 x H21 cm, which was used for this experiment. On the right, the thoracentesis pad's exterior and interior views are shown, providing a better understanding of its anatomical structure. For these experiments, we filled the pleural zone with water, as demonstrated in the figure.

The setup also included a Franka Emika Panda robot (Franka Emika GmbH, Germany), a mobile ultrasound machine (SONON 300L Imaging System, Orca Medical Ltd, UK) with a probe of 78 x 229 x 38 mm, connected via Wi-Fi to an Android tablet. Real-time ultrasound images were streamed to a computer with an Intel i7/16GB processor and an Intel UHD Graphics 770 graphics card using the serepy tool, as detailed in Section III-B.

B. Experiments in lab conditions

The autonomous thoracentesis system was evaluated through five trials in a controlled lab environment. During each iteration of the experiment, the robot performs an

autonomous thoracentesis cycle that begins with positioning the needle in the pleural zone using real-time ultrasound guidance. Once the needle is correctly placed, fluid is aspirated into the syringe, after which the valve system redirects the collected fluid to a drainage bottle. The cycle is repeated three times to extract a total of approximately 150ml of fluid, with the system ensuring precise synchronization between syringe retraction, fluid collection, valve repositioning, and fluid expulsion. Figure 8 presents four snapshots from the experiments, illustrating the critical stages of the procedure: US guidance, needle insertion, fluid drainage and aspiration

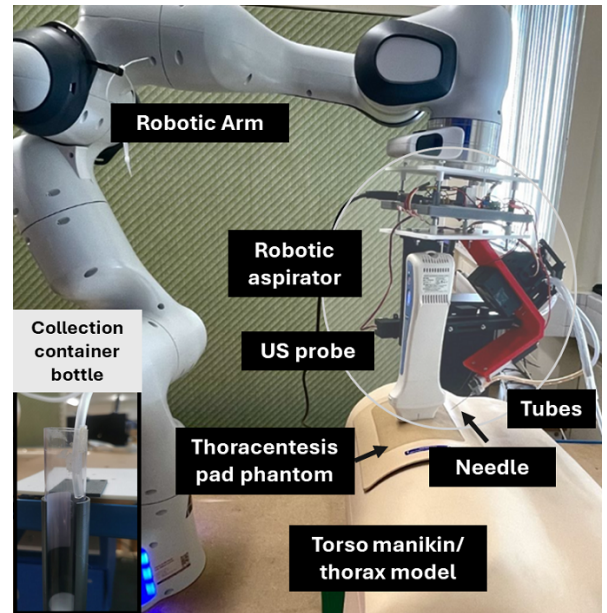


Fig. 6: Experimental setup.

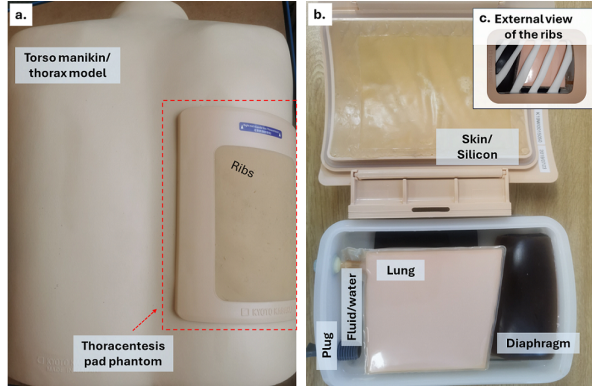


Fig. 7: Phantom views: a. torso manikin and thoracentesis pad phantom, b. interior structure of the thoracentesis pad, and c. its external view with the ribs.

to the collection bottle. In all the experiments, water used for the fluid aspiration process.

To evaluate aspirator's performance, key performance metrics adopted to capture the time taken for each stage of the procedure, the accuracy of pleural zone detection, the precision of needle insertion, and the volume of fluid aspirated. These metrics assess the system's consistency, reliability, and ability to perform autonomous thoracentesis with no human intervention. The quantitative results, including timings, detection accuracy, insertion precision, and fluid volume, are detailed in Table I, which presents the mean, standard deviation, and range (difference between minimum and maximum values).

Ultrasound Guidance and Monitoring. The robotic system successfully approached the thoracentesis pad and captured US images in real time. The system took an average of 30 seconds to scan and locate the pleural zone using image processing techniques. No latency was recorded for the processing of the images. Detection accuracy was calculated by comparing the detected pleural zone to the manually labeled zone across 100 frames. Accuracy was determined by measuring the overlap between the detected and actual zones, yielding a 95.6% match.

Needle Insertion Module (NIM). The NIM performance was evaluated by measuring the average needle insertion time (6.5 seconds) and the insertion accuracy in millimeters, based on the final position of the needle tip. The average deviation of the needle tip was less than 1 mm, which, compared to the pleural zone's average diameter of 5 mm, demonstrates a negligible error and high precision in all experiments.

Fluid Aspiration and Drainage Module (FADM). The fluid aspiration phase extracted on average 110 ml of fluid, within the target range of 100-150ml. The system completed the aspiration and drainage processes in 3 FADM cycles without any complications. In the end, the system retracted the needle, taking an average of 4.7 seconds.

Notable, the aspirator completed the procedure successfully in all experiments, resulting in an overall system accuracy of 100%. It is important to emphasize that the

TABLE I: Quantitative results from the experiments

Metric	Mean	Std	Range
Time for US Guidance (s)	30.0	2.5	25-34
Detection Accuracy (%)	95.6	2.4	92-98
Needle Insertion Time (s)	6.5	0.5	6-7
Needle Insertion Accuracy (mm)	1.3	0.6	1.0-2.0
Fluid Aspiration Volume (ml)	110	20	90-140
Needle Retraction Time (s)	4.7	0.3	4-5

results were obtained using the specific hardware, needles, syringes, and actuators available. Any modifications to the hardware could lead to increased fluid extraction and a faster process, highlighting the flexibility and potential for further optimization of the system.

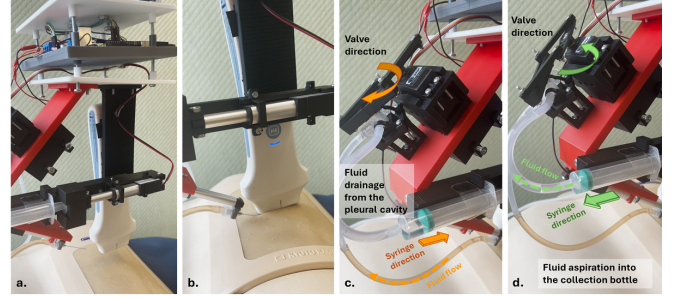


Fig. 8: Indicative snapshots from the experiments: a. starting pose; b. needle insertion; c. fluid drainage and fluid flow (orange); d. fluid aspiration into the collection bottle (green).

V. CONCLUSIONS

This paper presented the development of an autonomous, ultrasound-guided thoracentesis system featuring a novel robotic end-effector that integrates the US probe holder, the needle insertion mechanism and the fluid aspiration and drainage system. Extensive experiments demonstrated the system's reliability, with high accuracy in pleural zone detection (95.6%) and precise needle insertion (1 mm average error). The fluid aspiration process consistently removed fluid within the target range (1.2–1.35 L), and the system successfully operated with minimal variance across five trials.

Key achievements of this work include the successful integration of imaging and mechanical components, resulting in an automated, compact and modular solution that minimizes the risks associated with manual needle manipulation during thoracentesis. The system effectively combines engineering principles with medical requirements to deliver a reliable and precise robotic-assisted intervention.

For future work, several directions are proposed. First, optimizing the design with improved materials could create a more refined structure, enhancing both usability and patient comfort. Additionally, the implementation of deep learning algorithms for real-time pleural zone detection is a promising avenue to further enhance the system's accuracy and responsiveness. These advancements would not only increase the efficiency of the procedure but also broaden the system's applicability in various clinical scenarios.

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